

ccint — Progress Brief

Canada Cyber Social Intelligence: design rationale, build, and where it stands

23 September 2026

The question, and what changed about it

The project began as a longitudinal study: *how does Canada-related cybersecurity discussion change over time on social media?* A collector runs continuously against Bluesky, posts are labelled under versioned schemes, and change is tested statistically.

The first full analysis returned a negative result, and returned it with enough rigour to be worth something. The largest topic movement in the 30-day window, `data_breach` at +22%, has $p_{\text{FWER}} = 0.734$ under an author-level circular-shift null that preserves each author’s volume, topic mix and burstiness while destroying calendar alignment. Empirical power analysis — measured by author-pool replication rather than assumed — puts the requirement at roughly $8\times$ the current volume.

That result is only interesting if one knows what produced it. Three explanations were open, and nothing in the system could separate them, because every figure it produced came from labels it had produced itself. The work of the last phase was to build the measurements that can separate them. They do, and the answer was not the expected one: **the shortfall is not in the statistics and not mainly in the labelling — it is that the platform carries almost no human discussion of these events at all.**

Why the system is built this way

Four commitments shaped every design decision. Each exists to prevent a specific failure that would be invisible after the fact.

Bluesky, and one platform only. An open, documented API with usable historical search and terms that permit research collection; X and Reddit were excluded on access grounds rather than merit. The consequence is stated rather than hidden: this is a convenience sample, and the project’s job became partly to measure what it can show.

Labels are versioned and never overwritten. Annotation rows are keyed by (post, annotator version) and only ever inserted; four schemes coexist over the same corpus and diff row by row. Without this, every improvement to the annotator would silently rewrite history, and a shift in topic distribution could never be attributed to the world rather than to the method.

Raw data is kept verbatim; every number carries provenance. Payloads are stored unmodified, and every run and model label records what produced it, including the SHA-256 of the exact prompt text. That last part was added after noticing that a model annotator’s output is entirely determined by its prompt: “labels are never overwritten” had preserved the labels while losing the conditions that made them.

System failure must be distinguishable from social change. Every collection path records mode, status, window and counts, so a day with half the usual volume is attributable to either the world or a token expiry. The principle held under test: a 42-run token outage appears as 42 `partial` runs, not as a drop in Canadian cyber discussion.

Underlying all four: **measure before improving**. Repeatedly the intuitive next step turned out to be aimed at the wrong quantity, and only measurement revealed it — tuning the trend score would have been wasted, relaxing the relevance rules helps but not for the population that seemed to need it, and the most recent prompt improvement introduced a regression the evaluation set certified as correct.

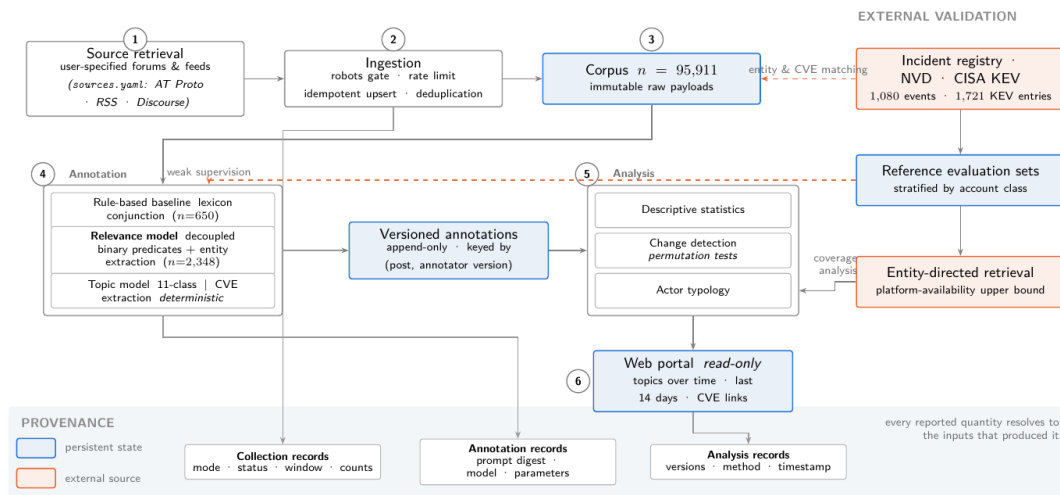


Figure 1: Processing pipeline. Stages 1–3 acquire and store the corpus, stage 4 annotates it under coexisting versioned schemes, stage 5 analyses the result. The external-validation column supplies labels no component of the pipeline produced.

How it was built

Collection runs in two modes against the fixed term list — a day-by-day backfill and an hourly incremental — with idempotent ingestion, stale-run reaping and client-timestamp repair; the corpus stands at 95,943 posts across three sources. Annotation proceeded in four generations, all retained: two lexicon-based, one that re-decides topic only while inheriting the relevance verdict verbatim so the two changes stay separately attributable, and one that replaces the relevance decision entirely. Analysis covers descriptive statistics, permutation testing against two null models, behavioural broadcast detection with a collection-lag control, and a two-axis actor typology separating measured audience from content-judged function. The most recent phases added the external-validation column of Figure 1 and then three capability layers: user-specified sources, deterministic CVE linking, and a read-only portal.

Sources are configuration, not code. `config/sources.yaml` names the forums and feeds to read; two generic collectors (RSS/Atom and the Discourse public JSON API) cover most of what a security community exposes. Both consult `robots.txt` and refuse a source they cannot read, because a collector that gets blocked produces a gap, and a gap is indistinguishable from a fall in discussion — the confusion this project exists to prevent. Each source must declare why automated access is believed permissible; the loader rejects one that does not. Web sources reuse the existing ingest path rather than adding a second, since a second path is a second way to bypass run bookkeeping.

Where it stands

The collection boundary was never measured

The 34 search terms came from a specification naming four examples followed by “etc.”; the remaining 26 were supplied during implementation from domain intuition, never derived from data and never revised. Reconstructing per-term productivity offline over the stored corpus:

Productivity is close to degenerate. `cybersecurity` returned 22,498 posts and 200 relevant ones; `ransomware` 8,881 and 105; `CVE` 12,502 and **three**. Eleven terms returned no relevant post at all, and one returned no post at all. `CVE` accounts for 13.7% of everything collected; `cybersecurity` alone accounts for 30.8% of all relevant posts, so the corpus time series is, to first order, one search term’s time series, and nothing monitors it. None of the 34 terms names Canada, so any Canadian cyber post not phrased with one of them is unreachable by construction.

Replacing the relevance gate

The old decision was a conjunction of two narrow lexicons, so recall was bounded by their intersection: 650 of 91,551 posts passed and the 89,714 rejected had never been examined by anything semantic. The replacement poses two *independent* predicates rather than one conjunction — a single combined prompt causes a 4B model to collapse the conjunction onto the side it knows better, admitting global threat-intelligence noise at 6.8% against a true Canadian rate of 1.1%.

Relevance gate	recall @ registry	specificity @ registry	corpus relevant
<code>rules_v2</code> — keyword conjunction	46.5%	99.8%	650
<code>llm_v2c</code> — two questions + validation	68.3%	99.4%	2,348

Of the 1,799 posts newly admitted, 84.0% had been rejected on the Canadian side of the conjunction. Two results here are method rather than number. Entity offsets are computed with `str.find` rather than requested from the model, on the narrow grounds that small models miscount them — which turned into a free hallucination detector, since an entity that cannot be located in the post was invented, and nearly half of extracted `place` entities failed it. And the residual error is neither a prompt nor a capability problem: the posts still missed carry no country marker at all, so closing it needs a gazetteer of Canadian organisations, not a larger model.

The external reference, and the central result

Until this phase every precision and recall figure traced to a single author who wrote the rules, the prompts, and the hand-coded evaluation set. A reported $\kappa = 0.870$ measured agreement under those conditions, not independence. Ingesting a public ransomware victim registry (1,080 events, 1,943 entities) supplies labels no part of the system produced.

For every Canadian event in the window, the victim organisation’s name was also searched directly on the platform, with no cybersecurity term attached, and every hit passed through the cyber judgement before counting:

Canadian ransomware events, 22 Aug – 22 Sep 2026	Events	Share
Listed in the registry	27	—
At least one cyber-related post exists on Bluesky	26	96%
At least one such post was collected by ccint	26	96%
At least one came from a non-feed account	2	7%

The 144 relevant posts come from 15 accounts. Twelve are automated threat-intelligence feeds; the three that are not are Canadian news organisations contributing five posts between them, all about a single event. Individual users contributed nothing; 94.9% of the posts have no likes.

What looks like coverage is syndication. The collector is not the bottleneck here — it captured 26 of the 26 events the platform carried. No ransomware trend is detectable because there is almost no ransomware *discussion* to detect: the corpus is a near-complete mirror of several leak-site feeds. That also reframes the power analysis — 8× of this corpus is 8× as much syndication, so volume is the wrong lever.

The one exception is instructive. The Commission de la construction du Québec, listed by the Qilin group with roughly 350,000 records taken, drew genuine Canadian media coverage in both languages. Four of the five posts are in the corpus; the fifth, a radio interview with the CCQ’s chief executive, carries no term from the 34-term list because *protection préventive* is in no lexicon — the collection-boundary gap at human scale.

Linking discussion to vulnerabilities

CVE identifiers are extracted by regular expression, not by a model. The format is fixed by MITRE, so the extraction is deterministic — which makes this the one table in the system whose trustworthiness does not depend on annotation quality. Identifiers are then enriched from two independent external sources: NVD supplies theoretical severity, and the CISA Known Exploited Vulnerabilities catalogue records that exploitation has actually been observed. The two are kept in separate columns and deliberately never combined into a single risk score: one is a judgement about how bad a flaw could be, the other is an observation that someone is using it, and merging them would conceal which part is external fact.

Across the corpus, 10,820 posts mention 5,404 distinct CVE identifiers. Only 147 of those — 2.7% — appear in the KEV catalogue. Those 147 attract 19.3% of all mentions, a concentration of roughly 7×, and 22 of the 25 most-discussed identifiers are among them.

Attention tracks exploitation, not severity. This is the first result in the project where social-media attention aligns with an external ground truth rather than merely mirroring a feed. It is also a usable one: a monitor that ranks vulnerabilities by how much they are being discussed recovers, without being told, most of the set that CISA has independently confirmed as exploited.

A methodological result about evaluation

The prompt revision that fixed entity recall simultaneously generalised “a .ca domain counts” into “a domain counts”, inflating the corpus from 906 to 3,528 posts with a third of them spurious — while the external negative set reported 99.0% specificity throughout, because it is built from formulaic English feed posts about foreign victims and contains none of the model’s actual failure modes. **An external label removes self-reference but does not confer representativeness, and a non-representative evaluation set does not merely fail to catch a regression — it certifies one.** Two deterministic checks catch it at no cost; the accompanying iteration report gives the full account.

What the project is, and what comes next

The original framing — trend analysis of Canadian cyber discourse — is not available at this volume for this event class, and adding data will not make it so. What is available, and appears to be unreported in the surveyed literature, is a sensor that states its own blind spots numerically. Published systems report how many events they detected (84 of 151 in the strongest comparable work); none reports how many events had no platform presence at all, because establishing that requires the per-event entity-directed retrieval used here, affordable only when the event count is small. This project's greatest liability — a corpus of a few thousand relevant posts — is precisely what makes the denominator measurable.

Immediate priorities, in order:

1. **Widen the registry beyond ransomware** — advisories, KEV, Canadian media. Every figure above is conditioned on a source that sees only leak-site victims: the narrowest limitation in the work.
2. **Feed the registry back into the relevance decision.** The residual failure is missing knowledge of Canadian organisation names, and the registry is a list of exactly those names.
3. **Add a Canada-side query stream** as an additional disjunctive stream, not a filter, to reach the population the registry cannot see.
4. **Have an independent annotator label the 200-post sample** already drawn and left blank — the one step that cannot be automated.
5. **Let the new sources accumulate.** Multi-source collection works but is not yet demonstrated at scale: the two feeds added so far contributed 45 posts, because RSS exposes only recent items and cannot be backfilled. A comparable time series requires either running them forward for weeks or adding a paginated source such as a Discourse forum.

Corpus 95,943 posts across three sources, 22 Aug – 23 Sep 2026; registry and KEV catalogue retrieved 22–23 Sep 2026; five coexisting annotation versions; 252 tests passing. Methodology in the project wrap-up, coverage measurements in the iteration report.